EXPERIMENT: TRAVELLING SALESMAN PROBLEM

Aim

To write a Python program to implement the Travelling Salesman Problem (TSP).

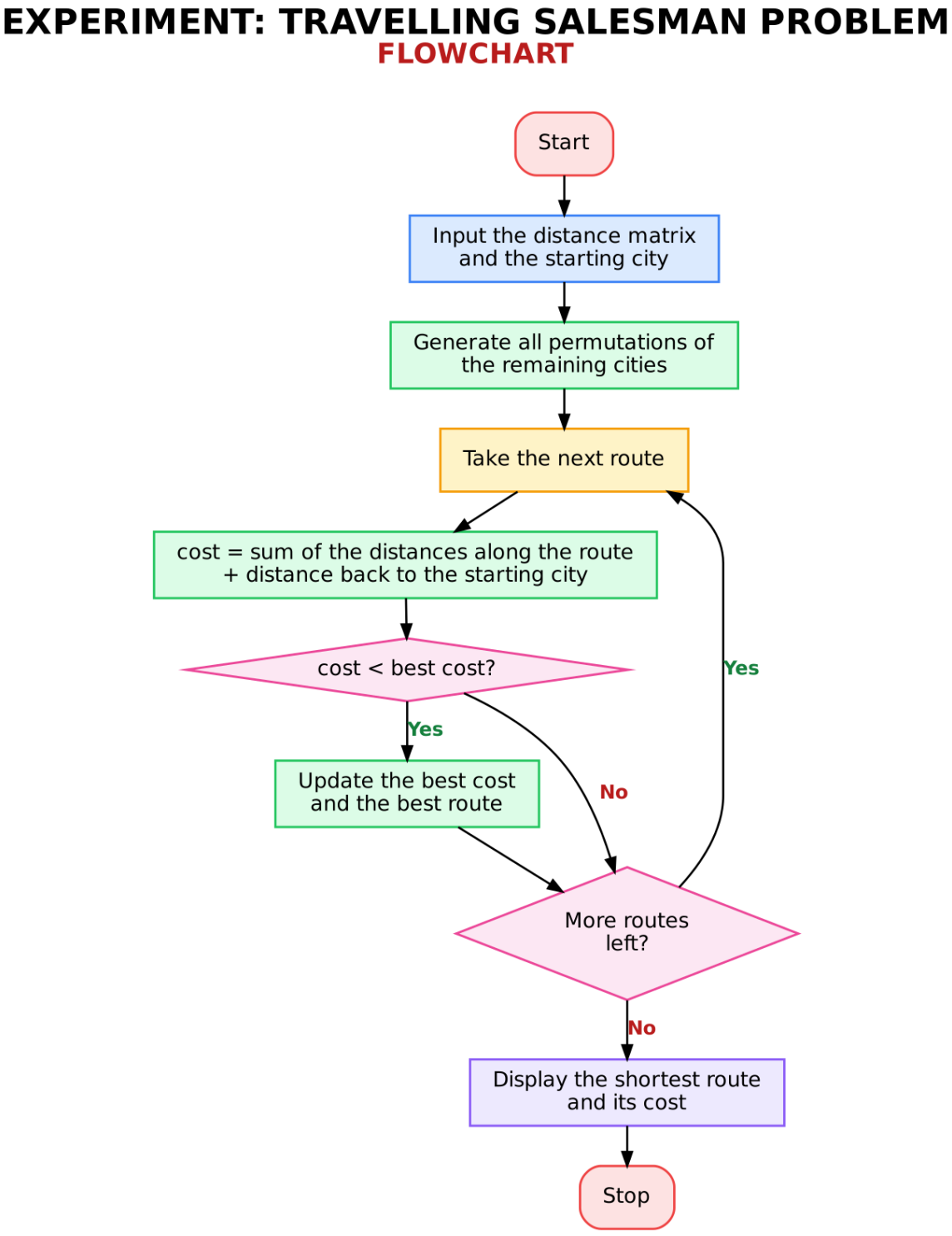
Objective

- To understand the Travelling Salesman Problem as an optimization problem.
- To implement an exhaustive (brute-force) search over all possible routes.
- To find the shortest route that visits every city exactly once and returns to the starting city.

Algorithm

1. Start.
2. Define the distance matrix of the cities and choose the starting city.
3. Generate all possible permutations of the remaining cities.
4. For each permutation, calculate the total distance of the route, including the return to the starting city.
5. If the total distance is less than the best distance found so far, store it as the best route.
6. Repeat steps 4 and 5 for all permutations.
7. Display the shortest route and its cost.
8. Stop.

Flowchart:



Code (Python)

```
from itertools import permutations

def travelling_salesman(dist, start=0):
    n = len(dist)
    cities = [i for i in range(n) if i != start]
    best_cost = float("inf")
```

```
    best_path = None

    for perm in permutations(cities):
        cost = 0
        current = start
        for city in perm:
            cost += dist[current][city]
            current = city
        cost += dist[current][start]          # return to the starting city

        if cost < best_cost:
            best_cost = cost
            best_path = (start,) + perm + (start,)

    return best_path, best_cost

# Distance matrix for 4 cities (0, 1, 2, 3)
dist = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

path, cost = travelling_salesman(dist)

print("Shortest route:", " -> ".join(str(c) for c in path))
print("Minimum cost  :", cost)
```

Output

Python Shell - Output

```
Shortest route: 0 -> 1 -> 3 -> 2 -> 0
Minimum cost  : 80
```

Output Result

Thus, the Python program to implement the Travelling Salesman Problem was executed successfully. The shortest route is 0 -> 1 -> 3 -> 2 -> 0 with a minimum cost of 80.

Conclusion

The experiment helped in understanding optimization problems. The Travelling Salesman Problem was solved by checking every possible route, which gives the optimal answer but grows very quickly (n!) as the number of cities increases.